



The 8th Annual

Parallel Applications Workshop Alternatives to MPI+X

Sunday, November 16, 2025
9:00 am - 5:30 pm (CST)

Hybrid Event

Held in conjunction with SC25
The International Conference for High Performance Computing, Networking, Storage, and Analysis





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Panel Discussion

**The Role of Alternatives to MPI+X Technologies
in AI/ML**

Format

Introduction to the panelists

Audience discussion on introductory statements

Moderator questions

Audience questions

Takeaway message from the panelists



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Pete Mendygral

- Former performance engineer for Cray Programming Environments
 - Focused on MPI, SHMEM, compilers, and AI
 - Application optimizations using MPI-RMA and non-blocking collectives
 - Developed “Cray DL Plugin” for full-system CNN training on Cori (CosmoFlow)
- Currently technical lead for DragonHPC
 - Runtime system for scalable Python workflows
 - AI+HPC
 - Data processing
 - Training and inference
 - Addresses the need for close coupling between application domains and/or parallel programming models
- MPI is great for ModSim and is otherwise useless for modern workflows
 - Just as the Earth is not the center of the universe, MPI is not the center of the HPC (let alone AI) universe
 - Maybe this is exactly how it should be

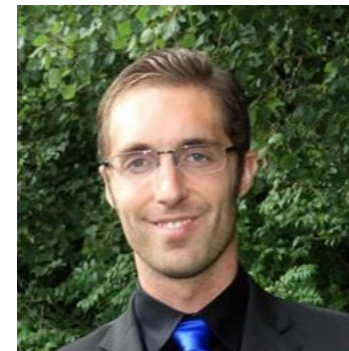


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Jeremiah Wilke, software engineer, CUDA software
at NVIDIA



Main work:

- Legate/Realm distribute runtime systems
- Improving CUDA developer experience in multi-node applications

General interests:

- Single-controller frameworks
 - Jax + Pathways
 - PyTorch + Monarch
- Graph compilers (XLA)
- Pipeline parallelism

Disclaimer: opinions presented here are my own and not necessarily those of NVIDIA



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Scientific computing could do more single-controller, but will remain multi-controller and MPI-focused

MPMD workflows in ML create complicated patterns that are best managed by a single controller

- Reinforcement learning

- Batch inference -> value function -> gradients -> optimizer step

- Pipeline parallelism

- Arbitrary interleaving of stages/microbatches (1F1B/Gpipe/Zero-bubble)

- Inference

- Scale up/scale down (elasticity) with changing demand, disaggregated prefill vs. decode

Argument:

- Aside from some viz/analysis or combined physics + DL, HPC rarely has interesting MPMD workflows
- Many HPC programs exist today in Jax, but MPI with checkpoint-restart works fine for majority of applications
- Unstructured or adaptive applications require special care to eliminate the coordinator as the bottleneck since partitioning is data-dependent
- Big AI companies will move mountains for 1-5% perf gains. HPC will not.



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Zhihao Jia

- Assistant Professor, CMU CS
- Research: ML systems and compilers, parallel computing
- Current focus: large-scale, high-performance systems for ML
- Projects: Mirage, FlexFlow, Legion